Effect of frequency-to-place mismatch and frequency warp on speech and music sound quality in acoustic cochlear implant simulation

Louis Villejoubert[1], Lorenzo Picinali[2], Kathleen Faulkner[3], Deborah Vickers[1]

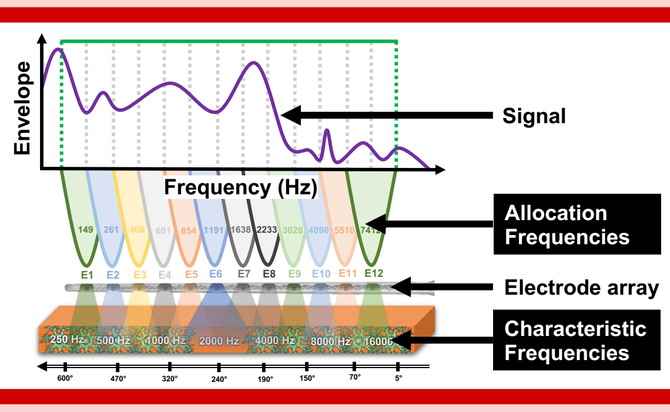
[1] Sound Lab, Cambridge Hearing Group, Department of Clinical Neuroscience, University of Cambridge, Cambridge, UK
[2] Dyson School of Design Engineering, Imperial College London, London, UK
[3] Oticon A/S, Smørum, Denmark

INTRODUCTION

In more recent years a greater emphasis has been placed on improving the **sound quality** for individuals using cochlear implants (CIs). Indeed, many post-lingually deafened adults with CIs have speech recognition performance close to normal hearing (NH) listeners in quiet environments, but yet complain about poor sound quality, especially for music. The contribution of CI technology to the **degradation of music quality** has been well documented, but little is known about the relative importance of different aspects relating to the technology and how they interact with the personal preferences and hearing characteristics of the individual.

INSERTION DEPTH → FTPM → DECREASE OF SPEECH UNDERSTANDING

One individual factor that is suspected to contribute largely to the sound quality degradation is the **frequency-to-place mismatch (FTPM)** induced by the **incomplete insertion of the electrode array** inside the cochlea. FTPM is defined as a deviation between the **characteristic frequency** and the **allocation frequency** assigned during CI fitting.



STUDY 1- EFFECT OF INSERTION DEPTH / FTPM ON SOUND QUALITY

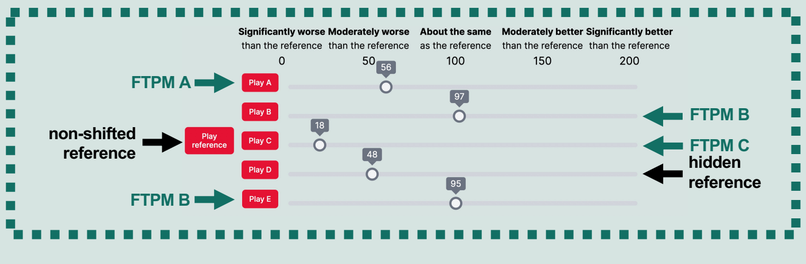
FTPM PROFILES → NOISE VOCODER ← SOUND MATERIAL → MUSHRA TEST → CORRELATION FTPM / SOUND QUALITY → RESEARCH QUESTION I: HOW FTPM IMPACTS SOUND QUALITY ?
SOUND MATERIAL → FEATURES → EXPLORATORY FACTOR ANALYSIS → RESEARCH QUESTION II: WHY SOME SOUNDS ARE MORE IMPACTED ?

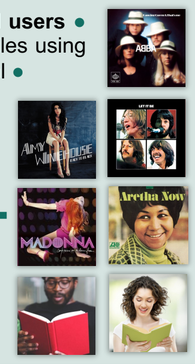
RATIONALE

Whereas much of the speech perception improvement occurs during the first months after CI activation, recent studies have shown that a significant FTPM could negatively affect this adaptation process. Alongside the improvements in speech perception over time from device activation patients often also change their reports of sound quality, from “high pitched”, “screechy”, “dalek-like” to more “normal” or “natural” after extensive listening experience. **However, the specific contribution of FTPM on sound quality isn’t clear yet.**

METHOD

• **8** NH listeners tested • FTPM from Canfarotta et al. (2022)[1], which were extracted from **real CI users** • Simulated using a **noise vocoder**[2] • The sound quality was evaluated for the different FTPM profiles using **MUSHRA test** • **Different sound materials** were tested: speech, music original, music instrumental •

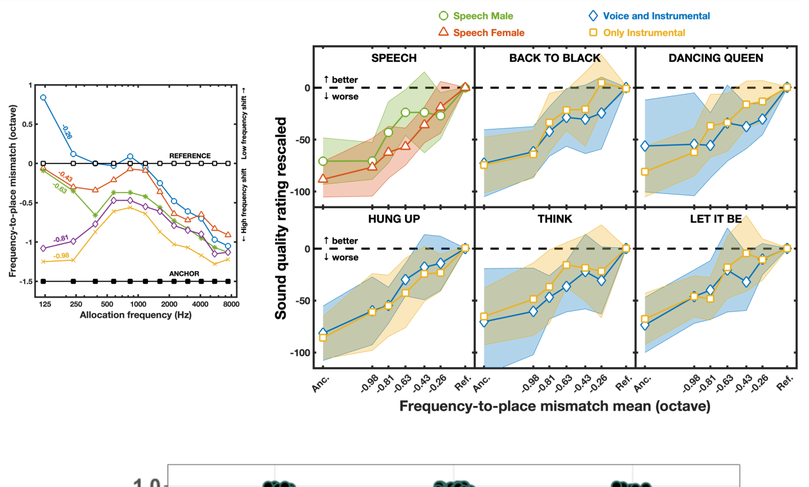




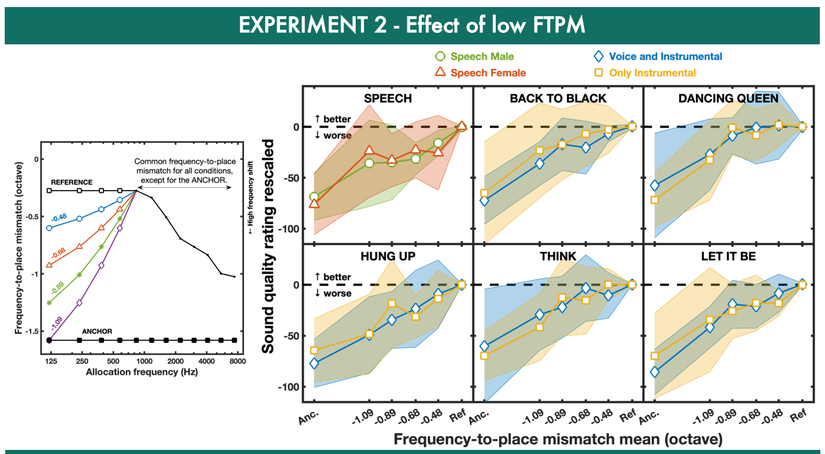
HOW FTPM IMPACTS SOUND QUALITY ?

Three experiments were conducted. For each experiment, different FTPM profiles were presented to the participant in the frame of MUSHRA interface. Such interface allows the participant to listen to a reference – here a **non-shifted reference** – and signals with FTPMs. The participant had the possibility to rank the sound quality of the sound in comparison to this reference. The scale was between 0 and 200 (rescaled between -100 and 100 in the results), with the possibilities to rank the sound quality as better (**0 → 100**), equal (**0**) or worse (**-100 → 100**) than the reference. **The correlations between FTPM means and scores** were calculated without the reference and the anchor, to specifically assess the effect of FTPM.

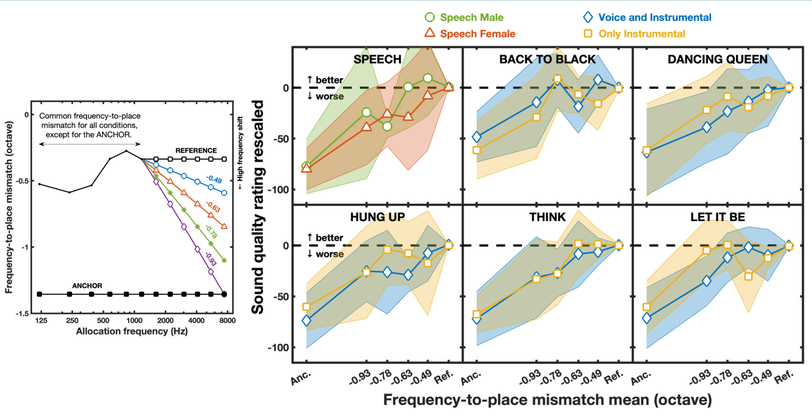
EXPERIMENT 1 - Effect of realistic FTPM profiles

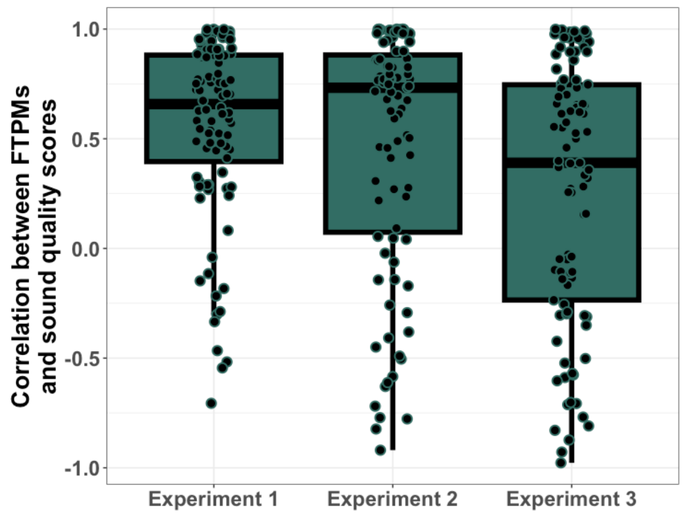


EXPERIMENT 2 - Effect of low FTPM



EXPERIMENT 3 - Effect of high FTPM





RESULTS AT A GLANCE

- Sound quality decreases with FTPM
- Sound quality is more impacted when FTPM is in the low frequencies
- Sounds interact differently with FTPM in terms of sound quality

WHY SOME SOUNDS ARE MORE IMPACTED ?

The correlation analysis of each sound clearly highlights the fact that each sound interacts differently with FTPMs. **The goal of this analysis is to understand which features of sound can explain the interaction between sound quality and FTPMs.** To this end, correlation indices and features have been examined/confronted for the 12 sounds tested.

STEP 1 - Features

The MIR toolbox[3] was used to extract features. More than 300 features were extracted, such as temporal, frequency, cepstral or modulation frequency features.

STEP 2 – Exploratory Factor Analysis (EFA)

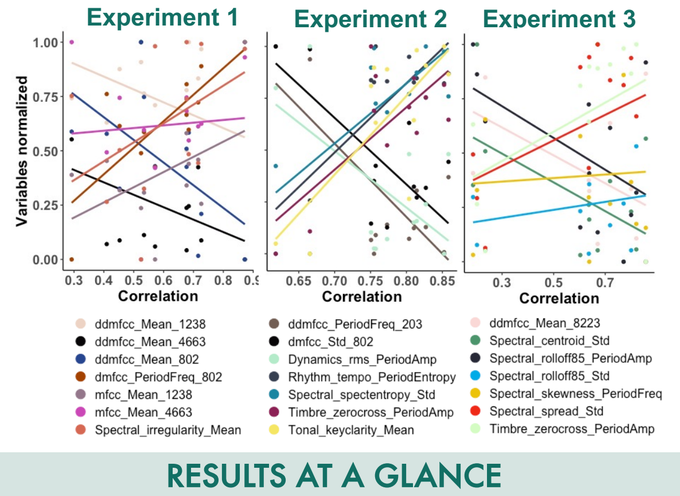
One exploratory analysis was done per experiment. The features selected for each analysis were different, restricted to low frequencies for the experiment 2 and high frequencies for the experiment 3. For each analysis, 11 latent variables (LVs) were collected.

STEP 3 – Correlation between Correlation Indices and LVs

	LV1	LV2	LV3	LV4	LV5	LV6	LV7	LV8	LV9	LV10	LV11
Experiment 1	0.21	0.35	0.4	-0.25	-0.25	-0.01	-0.52	0.06	-0.3	-0.4	-0.15
Experiment 2	-0.78	0.26	0.2	-0.04	0.09	-0.11	0.14	0.08	0.44	-0.13	-0.18
Experiment 3	0.03	0.13	0.39	-0.56	0.16	-0.17	0.08	0.53	-0.39	-0.09	-0.15

STEP 4 – Selection of features

The careful study of the variables contained in each LV did not allow us to classify it under a common categorization. Therefore, only a few variables were selected and reproduced below.



RESULTS AT A GLANCE

These features interact greatly with FTPM:

- **Mel-frequency cepstral coefficients (mfccs)**
- **Dynamic features (dmfccs and ddmfccs)**
- **Spectral content**

STUDY 2 – PREDICTING THE DEGREE OF FTPM USING PRE-SHIFTED SOUND

SPEECH & MUSIC → DIFFERENT LINEAR FREQUENCY SHIFT → FTPM profile → NOISE VOCODER → MUSHRA TEST → EXTRACTION OF THE LINEAR FREQUENCY SHIFT WITH THE BEST SOUND QUALITY
FTPM profile → FTPM mean → CORRELATION *(see results)*

RATIONALE

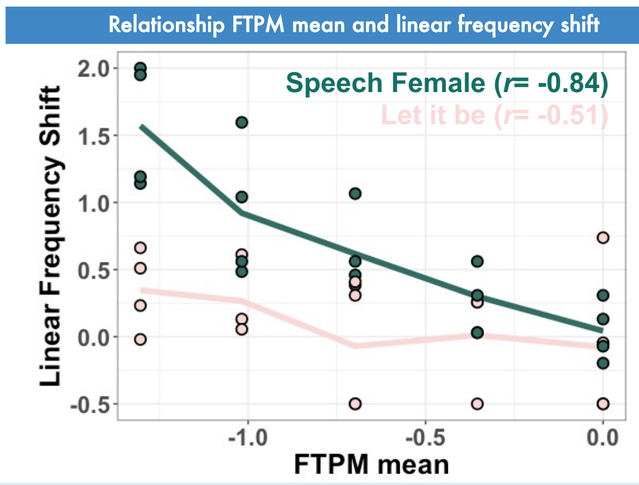
It is possible to investigate electrode placement (and FTPM) with imaging measures, such as multisection computed tomography (CT) scans, x-rays or conebeam CT. However, these measures are costly, their quality is not always precise enough to see cochlea structures or are just not always available clinically. Consequently, characterization of FTPM in clinics is rare. Without that information, the acclimatization process cannot be fully anticipated and there are no opportunities to intervene to maximise the adaptation process, if required.

This second study aims **to investigate FTPM by using a pre-frequency-shifted sound**. We assume that such manipulation may "counterbalance" the effect of FTPM. In other words, better sound quality should arise in such a case. **Hence, we should be able to relate the degree of FTPM with the frequency shift value that leads to the best sound quality**.

METHOD

• 4 NH participants have been tested • One speech (Speech Female) and one music (Let it be original) have been tested • 6 values of Linear Frequency Shift (LFS) - shifted in the low frequencies – were presented for each MUSHRA interface • A function has been fitted between LFSs and scores and the top of this function has been retained as the best LFS •

Relationship FTPM mean and linear frequency shift



RESULTS AT A GLANCE

• **“Speech Female” is able to predict the degree of FTPM** • **“Let it be” is less adapted for the prediction of FTPM** •

CONCLUSION

ACUTE SITUATION

FTPM ↘ → SOUND QUALITY ↗
FTPM low ↘ → SOUND QUALITY ↗
Shifted sound → FTPM

AFTER ADAPTATION..?

After the adaptation process with real CI users, what is the impact of FTPM?

FTPM ↘ ? → SOUND QUALITY ↗

What would be the explanation underlying the linear frequency shift chosen?

Shifted sound ? → FTPM

[1] Canfarotta, M. W., Dillon, M. T., Buss, E., Pillsbury, H. C., Brown, K. D., & O'Connell, B. P. (2020). Frequency-to-Place Mismatch: Characterizing Variability and the Influence on Speech Perception Outcomes in Cochlear Implant Recipients. Ear and hearing, 41(5), 1349–1361.
[2] Ben Williges, & Tim Jürgens. (2019). Pulsatile Cochlear Implant Vocoder v 1.0.1 (v1.0.1). Zenodo.
[3] Olivier Lartillot, Petri Toiviainen, “A Matlab Toolbox for Musical Feature Extraction From Audio”, International Conference on Digital Audio Effects, Bordeaux, 2007.

This work was done as part of the PhD project “**Characterizing and alleviating distortions in sound and music quality in adults using cochlear implants: a music re-engineering approach**”. This project was funded by the William Demant Foundation.